

49 Nervous Systems

KEY CONCEPTS

- 49.1** Nervous systems consist of circuits of neurons and supporting cells *p. 1086*
- 49.2** The vertebrate brain is regionally specialized *p. 1091*
- 49.3** The cerebral cortex controls voluntary movement and cognitive functions *p. 1096*
- 49.4** Changes in synaptic connections underlie memory and learning *p. 1099*
- 49.5** Many nervous system disorders can now be explained in molecular terms *p. 1102*

Study Tip

Think in pairs: Regional specialization involves many examples of paired structures or circuits. Fill in the complementary or reciprocal functions of these pairs to help you understand their roles in the brain or nervous system.

Structure A/Function	Structure B/Function
CNS (brain and spinal cord) <i>Integration of information</i>	PNS (ganglia and peripheral nerves) <i>Transfer of information to/from CNS</i>
Sympathetic division of autonomic nervous system	Parasympathetic division of autonomic nervous system
Left hemisphere of brain	Right hemisphere of brain
Hippocampus (role in memory)	Cerebral cortex (role in memory)

Go to Mastering Biology

For Students (in eText and Study Area)

- Get Ready for Chapter 49
- The Visual Brain: Nervous System
- HHMI Video: The Human Suprachiasmatic Nucleus

For Instructors to Assign (in Item Library)

- Scientific Skills Exercise: Designing an Experiment Using Genetic Mutants
- Tutorial: The Vertebrate Nervous System

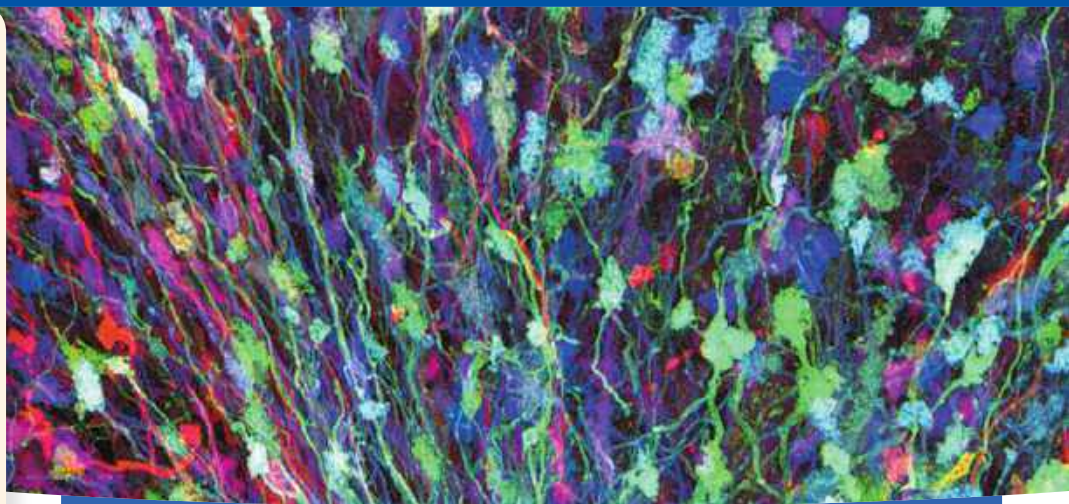
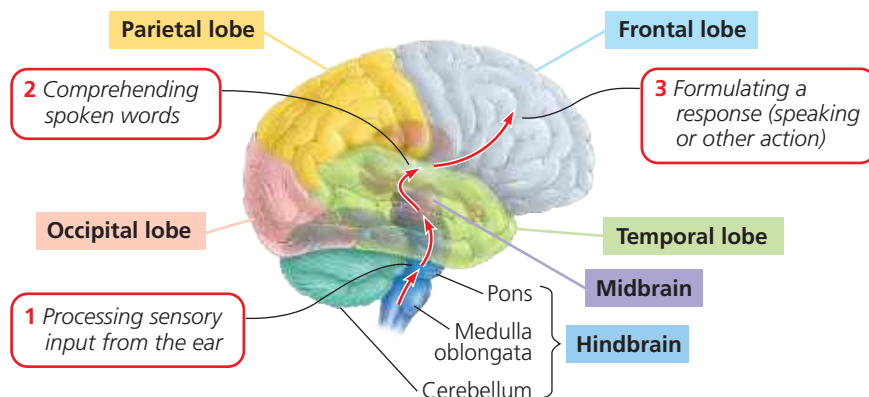


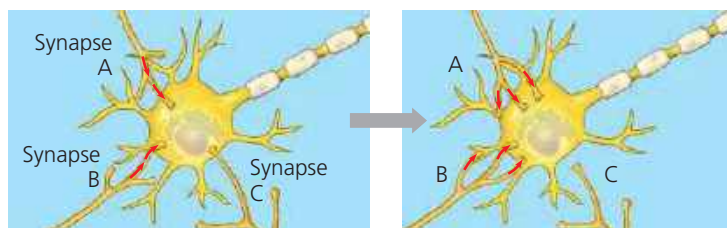
Figure 49.1 Neuroscientists engineered mice to express a random combination of four fluorescent proteins in each brain cell. The result, shown here, is a “rainbow,” with each neuron displaying one of 90 different color combinations. This rainbow technology holds promise for studying particular pathways within the mouse brain. Ultimately, however, we want to understand our own brains, which contain an estimated 10^{11} (100 billion) neurons and 10^{14} (100 trillion) connections.

How are billions of neurons organized to perform complex tasks?

Regional specialization: Complex tasks, such as responding to a spoken question, involve the stepwise functions of different brain regions.



Memory formation: Information is stored by reinforcing patterns of active connections (synapses) among particular neurons.



Synapses that are active in synchrony (A and B) are strengthened. Synapses that are not part of an active circuit (C) are weakened or lost.